

Curso:

1. Utilizar la técnica de mínimos cuadrados para ajustar los siguientes datos

x	-2	-1	0	1	2
y	-3	-1	0	3	5

mediante una función de la forma $y = ax + b$.

2. Determinar el conjunto de todos los $x \in \mathbb{R}^3$ tales que $A^{1961}x = [2 \ 0 \ 2]^T$, donde

$$a(-2) + b = -3$$

$$a(-1) + b = -1$$

$$a(0) + b = 0$$

$$a(1) + b = 3$$

$$a(2) + b = 5$$

$$-2a + b = -3$$

$$-a + b = -1$$

$$b = 0$$

$$a + b = 3$$

$$2a + b = 5$$

$$\begin{pmatrix} -2 & 1 \\ -1 & 1 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -3 \\ -1 \\ 0 \\ 3 \\ 5 \end{pmatrix}$$

$$\text{Aplico } A^T A \hat{x} = A^T b$$

$$\begin{pmatrix} -2 & -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -2 & 1 \\ -1 & 1 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -2 & -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -3 \\ -1 \\ 0 \\ 3 \\ 5 \end{pmatrix}$$

$$\begin{pmatrix} 10 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} \hat{a} \\ \hat{b} \end{pmatrix} = \begin{pmatrix} 20 \\ 4 \end{pmatrix}$$

$$\begin{cases} 10\hat{a} = 20 \\ 5\hat{b} = 4 \end{cases} \rightarrow \begin{cases} \hat{a} = 2 \\ \hat{b} = \frac{4}{5} \end{cases}$$

$$y = 2x + \frac{4}{5}$$

2. Determinar el conjunto de todos los $x \in \mathbb{R}^3$ tales que $A^{1961}x = \begin{bmatrix} 2 & 0 & 2 \end{bmatrix}^T$, donde

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}.$$

Nota

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

I) calculo los autovalores

$$p_A(\lambda) = \det \begin{pmatrix} 1-\lambda & 1 & 0 \\ 1 & 2-\lambda & -1 \\ 0 & -1 & 1-\lambda \end{pmatrix}$$

$$= (1-\lambda)^2(2-\lambda) - [(1-\lambda)(1-\lambda)]$$

$$= (1-\lambda)(2-\lambda)(1-\lambda) - [0 + (1-\lambda) + (1-\lambda)]$$

$$= (1-\lambda)^2(2-\lambda) - 2(1-\lambda)$$

$$= (1-\lambda)[(1-\lambda)(2-\lambda) - 2]$$

$$= (1-\lambda)[2 - \lambda - 2\lambda + \lambda^2 - 2]$$

$$= (1-\lambda)[\lambda^2 - 3\lambda]$$

$$= (1-\lambda)[\lambda^2 - 3\lambda]$$

$$= (1-\lambda)\lambda(\lambda - 3)$$

$$\boxed{\lambda_1 = 1}$$

$$\boxed{\lambda_2 = 0}$$

$$\boxed{\lambda_3 = 3}$$

Busco los autoespacios de cada autovalor

$$S_1 = \text{Nul} \left(\begin{pmatrix} 1-1 & 1 & 0 \\ 1 & 2-1 & -1 \\ 0 & -1 & 1-1 \end{pmatrix} \right) \rightsquigarrow \text{Nul} \left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & -1 \\ 0 & -1 & 0 \end{pmatrix} \right)$$

$$\text{Nul} \left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \right)$$

$$x_2 = 0$$

$$x_1 + x_2 - x_3 = 0$$

$$x_1 - x_3 = 0 \Rightarrow x_1 = x_3$$

$$\begin{aligned} [x_1 \ x_2 \ x_3]^T &= [x_1 \ 0 \ x_1]^T \\ &= x_1 [1 \ 0 \ 1]^T \end{aligned}$$

$$\boxed{\$1 = \text{gen} \{ [1 \ 0 \ 1]^T \}}$$

$$\text{Busco } \$0 = \text{Nul} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} \sim \text{Nul} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\begin{cases} x_1 + x_2 = 0 \\ x_2 - x_3 = 0 \end{cases} \rightarrow \boxed{x_2 = -x_1} \\ -x_1 - x_3 = 0 \Rightarrow \boxed{x_3 = -x_1}$$

$$\begin{aligned} [x_1 \ x_2 \ x_3]^T &= [x_1 \ -x_1 \ -x_1]^T \\ &= x_1 [1 \ -1 \ -1]^T \end{aligned}$$

$$\boxed{\$0 = \text{gen} \{ [1 \ -1 \ -1]^T \}}$$

$$\begin{aligned} \text{Busco } \$3 &= \text{Nul} \begin{pmatrix} 1-3 & 1 & 0 \\ 1 & 2-3 & -1 \\ 0 & -1 & 1-3 \end{pmatrix} = \text{Nul} \begin{pmatrix} -2 & 1 & 0 \\ 1 & -1 & -1 \\ 0 & -1 & -2 \end{pmatrix} \\ &= \text{Nul} \begin{pmatrix} -2 & 1 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

$$\begin{cases} -2x_1 + x_2 = 0 \rightarrow x_2 = 2x_1 \\ -x_2 - 2x_3 = 0 \rightarrow -2x_1 - 2x_3 = 0 \\ 2x_3 = -2x_1 \\ \boxed{x_3 = -x_1} \end{cases}$$

$$[x_1 \ x_2 \ x_3] = [x_1 \ 2x_1 \ -x_1] = x_1 [1 \ 2 \ -1]$$

$$\boxed{\$3 = \text{gen} \{ [1 \ 2 \ -1]^T \}}$$

$$B = \left\{ [1 \ 0 \ 1]^T, [1 \ -1 \ -1]^T, [1 \ 2 \ -1]^T \right\}$$

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & -1 \end{pmatrix}$$

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$P = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ -1 & 1 & -1 \end{pmatrix}$$

$$D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ -1 & 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ -1 & 1 & -1 \end{pmatrix}^{-1}$$

Para calcular $A^{1961} x = [2 \ 0 \ 2]^T$

$$x = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$$

Aplico

$$A^{1961} x = \alpha_1 A^{1961} v_1 + \alpha_2 A^{1961} v_2 + \alpha_3 A^{1961} v_3$$

$$A^{1961} x = \alpha_1 (0)^{1961} v_1 + \alpha_2 (1)^{1961} v_2 + \alpha_3 (2)^{1961} v_3$$

$$A^{1961} x = \alpha_2 v_2 + \alpha_3 (2)^{1961} v_3$$

Sabemos que $A^{1961} x = [2 \ 0 \ 2]^T$

$$\alpha_2 v_2 + \alpha_3 (2)^{1961} v_3 = 0 v_1 + 2 v_2 + 0 v_3$$

$$\alpha_1 \in \mathbb{R}$$

$$\alpha_2 = 2$$

$$\alpha_3 = 0$$

La solución entonces es

$$x = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$$

$$x = \alpha_1 \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$x = \alpha_1 \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + 0 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$x = \alpha_1 \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} \quad \alpha_1 \in \mathbb{R}$$